

Mathematical overview

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The problem

A convex body $K \subset \mathbb{R}^3$ has constant width d when the distance between every pair of parallel supporting planes with opposite normals is d . The three-dimensional Blaschke-Lebesgue problem asks for the least possible volume among all such bodies.

The two classical Meissner bodies have volume

$$V_M(d) = \frac{\pi}{12} \left(8 - 3\sqrt{3} \arccos \frac{1}{3} \right) d^3 = 0.419860045965080 \dots d^3,$$

and are conjectured to minimize volume. That conjecture remains separate from the theorem proposed here.

The present package represents the universal lower bound

$$\text{Vol}(K) \geq \frac{26220940089713\pi}{20000000000000} d^3 = 0.4118775637803022742 \dots d^3.$$

Normalize the width

Volume scales cubically. It is therefore enough to prove the result for width one and multiply the answer by d^3 .

For a width-one body, the support function satisfies

$$h_K(u) + h_K(-u) = 1.$$

Equivalently,

$$K + (-K) = B(0, 1),$$

so the diameter is one. In three dimensions, Blaschke's identity gives

$$S(K) = 2 \operatorname{Vol}(K) + \frac{2\pi}{3}.$$

This converts a surface-area estimate into a volume estimate.

Minimum circumsphere and the circumradius split

Place the origin at the center of a minimum circumscribed ball $B(0, R)$. The contact points with the ball balance around the origin. Caratheodory's theorem reduces this balance to at most four contacts, and the diameter-one condition gives Jung's range

$$\frac{1}{2} \leq R \leq \frac{\sqrt{6}}{4}.$$

The proof splits this interval at the exact rational number

$$R_* = \frac{305695601}{500000000} = 0.611391202.$$

The two branches use different geometric information.

Branch I: lower circumradius

Suppose $R \leq R_*$. An outer support point x with unit normal n has an opposite point $x - n \in K$. Since $K \subset B(0, R)$,

$$\langle x, n \rangle \geq \frac{|x|^2 + 1 - R^2}{2}.$$

This inequality controls how the radial function ρ can change across the sphere.

A divergence estimate with

$$F_m(x) = \frac{2x}{|x|^2 + m}$$

gives

$$3V = \int_{\mathbb{S}^2} \rho^3 d\sigma, \quad S \leq \int_{\mathbb{S}^2} \frac{2\rho^3}{\rho^2 + m} d\sigma.$$

The integrand is majorized by a tangent line in the variable $X = \rho^3$. The nonnegative tangent gap can be recovered on spherical caps where the radial function is forced above or below explicit values.

All cap angles are represented by a rational tangent-half-angle grid. After one square root is cleared, every propagation, cap, disjointness, and integration condition becomes an integer inequality. A gap-free finite table covers $[1/2, R_*]$. Its unique weakest row is 03B02, and its exact coefficient of π is the final theorem coefficient.

Branch II: near the Jung endpoint

Suppose $R \geq R_*$. Four circumsphere contacts are required and are close to a regular tetrahedral configuration.

Let $p_i = Ru_i$ be the contact points and define the six edge deficiencies

$$e_{ij} = 1 - |p_i - p_j|^2 \geq 0, \quad S = \sum_{i < j} e_{ij}.$$

The weighted balance equations imply the scalar budget

$$S \leq \frac{3 - 8R^2}{4R^2 - 1}.$$

Because the right side decreases with R , the split radius gives one exact universal deficiency budget on the entire near-Jung branch.

Affine tetrahedral coordinates

Use the four sign vectors

$$(1, 1, 1), \quad (1, -1, -1), \quad (-1, 1, -1), \quad (-1, -1, 1)$$

as a regular tetrahedral reference. An affine map sends this reference configuration to the actual contact tetrahedron. Its normalized Gram matrix H is an explicit linear function of the six deficiencies.

The proof obtains the determinant transfer

$$\det H \geq 1 - \frac{S}{2} - \frac{S^2}{2}.$$

Thus any certified volume in reference coordinates transfers to physical coordinates with a controlled loss.

High-resolution forced fan

The four contacts and the diameter-one ball constraints force a radial polyhedral core. The canonical fan contains:

- 98,306 vertices;
- 196,608 triangular faces;
- 294,912 edges; and
- 24,833 orbits under the distinguished-edge stabilizer.

The fan rays carry exact integer radial data and dyadic S-procedure multipliers. The minimum exact fan margin is

$$2.8889042070928578767 \times 10^{-10} > 10^{-12}.$$

After determinant transfer, the forced core contributes at least

$$V_{\text{core}} \geq 0.4117357103127771735489695 \dots$$

Dominant-edge coordinate-width lemma

Relabel a maximum-deficiency edge as 01. The normalized Gram diagonal obeys

$$H_{00} \leq 1 + \frac{S}{2}, \quad H_{11}, H_{22} \leq 1 + \frac{S}{6}.$$

For a positive-definite matrix,

$$(H^{-1})_{jj} \geq \frac{1}{H_{jj}}.$$

These inequalities give three different lower bounds for the physical widths of transformed coordinate sections. The two transverse directions receive a stronger estimate than the dominant-edge direction. This anisotropic coordinate-width step is the main analytic sharpening that separates the present handoff from the predecessor certificate.

Six additional cone caps

For each coordinate axis, one cap lies on the positive side and one on the negative side. Exact section polygons, support values, cap depths, and allocation brackets give three separately optimized antipodal cap-pair lower bounds. Their sum is

$$V_{\text{caps}} \geq 0.00014396090205525818382845 \dots$$

An exact transformed-ellipsoid argument proves that the interiors of all six caps are pairwise disjoint, so their volumes may be added to the forced core.

Near-Jung assembly

The near-Jung branch yields

$$V_{\text{near}} \geq 0.41187967121483243173279798 \dots$$

This exceeds the upper enclosure of the lower-branch theorem value by

$$0.000002107434530157484619055 \dots > 0.$$

Therefore the universal bottleneck is the lower-branch row 03B02, and the two branches together cover the complete Jung interval.

What the computer checks

The exact verifier checks the finite rational objects produced after the mathematical reductions:

- lower-row propagation chains;
- fan-ray inequalities and multipliers;
- section hulls and exact areas;
- determinant and coordinate-width inputs;
- cap allocations and disjointness;
- branch handoff; and
- the exact final coefficient.

The mathematical derivation connecting arbitrary constant-width bodies to those finite conditions is written in the paper and remains subject to human review. The division is detailed in [../certificate/TRUST_BOUNDARY.md](#).

Reading path

A reviewer can proceed in this order:

1. [paper/contact_geometric_bound.pdf](#);
2. this overview;
3. [CLAIM_DEPENDENCIES.md](#);
4. [../certificate/CERTIFICATE_SPECIFICATION.md](#);
5. [APPENDIX_A_EXACT_CONSTANTS.md](#); and
6. the exact verifier and proof objects under `certificate/`.